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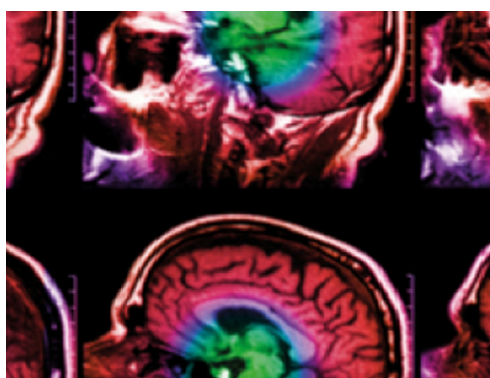
An attention-based deep learning model for predicting microvascular invasion of hepatocellular carcinoma using an intra-voxel incoherent motion model of diffusion-weighted magnetic resonance imaging

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PAPER

An attention-based deep learning model for predicting microvascular invasion of hepatocellular carcinoma using an intra-voxel incoherent motion model of diffusion-weighted magnetic resonance imaging

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17 September 2021Qingyuan Zeng^{1,3}, Baoer Liu^{2,3}, Yikai Xu^{2,*} and Wu Zhou^{1,*}¹ School of Medical Information Engineering, Guangzhou University of Chinese Medicine, People's Republic of China² Department of Medical Imaging Center, Nanfang Hospital, Southern Medical University, Guangzhou, People's Republic of China³ Equal contribution.

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E-mail: yikai@smu.edu.cn and zhouwu@gzcum.edu.cn**Keywords:** IVIM-DWI, attention, convolutional neural networks, hepatocellular carcinoma, microvascular invasion

Abstract

The intra-voxel incoherent motion model of diffusion-weighted magnetic resonance imaging (IVIM-DWI) with a series of images with different b -values has great potential as a tool for detecting, diagnosing, staging, and monitoring disease progression or the response to treatment. The current clinical tumour characterisation using IVIM-DWI is based on the parameter values derived from the IVIM model. On the one hand, the calculation accuracy of such parameter values is susceptible to deviations due to noise and motion; on the other hand, the performance of the parameter values is rather limited with respect to tumour characterisation. In this article, we propose a deep learning approach to directly extract spatiotemporal features from a series of b -value images of IVIM-DWI using a deep learning network for lesion characterisation. Specifically, we introduce an attention mechanism to select dominant features from specific b -values, channels, and spatial areas of the multiple b -value images for better lesion characterisation. The experimental results for clinical hepatocellular carcinoma (HCC) when using IVIM-DWI demonstrate the superiority of the proposed deep learning model for predicting the microvascular invasion (MVI) of HCC. In addition, the ablation study reflects the effectiveness of the attention mechanism for improving MVI prediction. We believe that the proposed model may be a useful tool for the lesion characterisation of IVIM-DWI in clinical practice.

1. Introduction

Diffusion-weighted imaging (DWI) has greatly improved in recent years, surpassing the original concept of the apparent diffusion coefficient (ADC). As a new technique, intra-voxel incoherent motion (IVIM) has entered the clinical field. IVIM diffusion magnetic resonance (MR) imaging has great potential as a tool for the treatment of cancer patients, and is widely used in the detection, diagnosis, staging, and monitoring of disease progression or the response to treatment (Iima and Le Bihan 2016). Hepatocellular carcinoma (HCC) is the most common primary malignant tumour in the liver, accounting for 75%–85% of all liver cancer cases (Sung *et al* 2021), and microvascular invasion (MVI) has been proven to be a key indicator of recurrence and poor prognosis (European Association for the Study of the Liver 2018, Yoh *et al* 2019). Knowing the likelihood of MVI before surgery can help determine treatment strategies and patient management (Erstad and Tanabe 2019). IVIM diffusion MR imaging has been explored to predict the MVI of HCC with highly variable results reported in the literature, and its utility has not been widely recognised due to the limitations of the IVIM model.

The IVIM model typically uses a bi-exponential function model to describe DWI data and a series of quantitative indicators that are generated using nonlinear least squares method fitting, including the water molecule diffusion rate, D ; pseudo-diffusion coefficient, D^* ; and tissue perfusion fraction, f (Le Bihan *et al* 1988). The clinical applications of IVIM-DWI are mainly based on the obtained IVIM parameter values for HCC characterisation (Woo *et al* 2014, Li *et al* 2018, Zhao *et al* 2018, Zhu *et al* 2018, Wei *et al* 2019a, 2019b). However, there are several limitations to the IVIM model used in current clinical applications. First, IVIM parameters are very sensitive to motion and artefacts in DWI, which significantly influences the lesion characterisation performance of IVIM-DWI. Accurate estimation of IVIM-related parameters requires fitting the DWI signal, which is often prone to errors and calculation-intensive (Koh *et al* 2011, Iima and Le Bihan 2016). Although effective fitting approaches have recently been proposed to obtain highly accurate IVIM parameters (Gustafsson *et al* 2017, Gurney-Champion *et al* 2018), they are still subject to estimation bias and image noise. Furthermore, accurate data fitting with models also requires the acquisition of multiple images with a large range of b -values, resulting in long acquisition times. A limited number of b -value images will inevitably influence the accuracy of parameter fitting and characterisation performance (Iima and Le Bihan 2016). More seriously, the voxel signal intensity of the IVIM parameter map is used as a feature in classification (Le Bihan *et al* 1988, Woo *et al* 2014, Li *et al* 2018, Zhao *et al* 2018, Zhu *et al* 2018, Wei *et al* 2019a, 2019b). Although it is very simple to use and can intuitively interpret clinical results, the characterisation performance based on a single feature is rather limited from the perspective of pattern recognition.

Deep features obtained using convolutional neural networks (CNNs) from data-driven learning have proven the superior performance of characterising lesion features in medical imaging processing (Greenspan *et al* 2016, Litjens *et al* 2017). A previous study concluded that deep features can be extracted directly from the ADC map of the DWI image (Christ *et al* 2017). Because heart motion and image artefacts can seriously affect the accuracy of the ADC map, their performance is still limited by the accuracy of the generated ADC map. Recently published research (Wang *et al* 2020) extracted the deep features of three b -value images obtained using DWI through multimodal fusion of deep convolutional networks, which could better predict the MVI of HCC. However, the number of b -values in IVIM imaging is far greater than that in conventional DWI, and the feature fusion method results in huge network parameters for training. Moreover, the relationship between adjacent b -value images has been ignored, and their role in lesion characterisation has not been explored. Inspired by breakthroughs in video processing for deep feature learning (Tran *et al* 2015), a series of b -value images of a tumour can be considered as a spatial and temporal dataset, and it is feasible to directly extract spatiotemporal features from a series of IVIM-DWI b -value images using a deep network for lesion characterisation. To this end, the relationship between adjacent b -value images can be investigated in the spatiotemporal domain for lesion characterisation. Furthermore, not all b -value images and deep features extracted from the hidden layers contribute to lesion characterisation, and worthless features potentially influence characterisation performance. Therefore, the attention mechanism (Vaswani *et al* 2017) was introduced to remove redundant features and focus on dominant features in the process of spatiotemporal feature extraction from a series of IVIM-DWI b -value images for performance improvement.

In this study, unlike extracting features from IVIM parameter maps (Christ *et al* 2017) or from each b -value image (Wang *et al* 2020) for lesion characterisation, we propose to extract spatiotemporal features directly from a series of IVIM-DWI b -value images using a deep learning network to characterise lesions. Specifically, an attention mechanism was adopted to select dominant features from specific b -values, network channels, and spatial areas for better lesion characterisation. Furthermore, by visualising the weight vector and matrix of the attention mechanism related to the lesion characteristics, it is helpful to understand and interpret the relationship between IVIM b -value images and pathological information with respect to lesions. We conducted extensive experiments on predicting the MVI of clinical HCCs using IVIM-DWI to demonstrate the effectiveness and superiority of the proposed method.

2. Methods

2.1. Study population, imaging protocol, and pathological information

This retrospective study was approved by the local institutional review board, and informed consent was not required. A total of 98 consecutive patients with 107 HCCs between January 2017 and September 2020 were included in the study; seven patients had two lesions, and the remaining 91 patients had one lesion each. Since multiple lesions in a patient may be inherently related, to maintain the independence of experimental samples, we only included the largest lesion from patients with multiple lesions in the analysis. Finally, 98 HCCs were included in this retrospective study. Routine IVIM-DWI serial examinations were performed for each patient using a 3.0-T MRI system (Achieva, Philips Healthcare, The Netherlands) as part of preoperative MR imaging. The IVIM-DWI sequence was performed in the axial plane (repetition time/echo time = 1973/57, 336×336

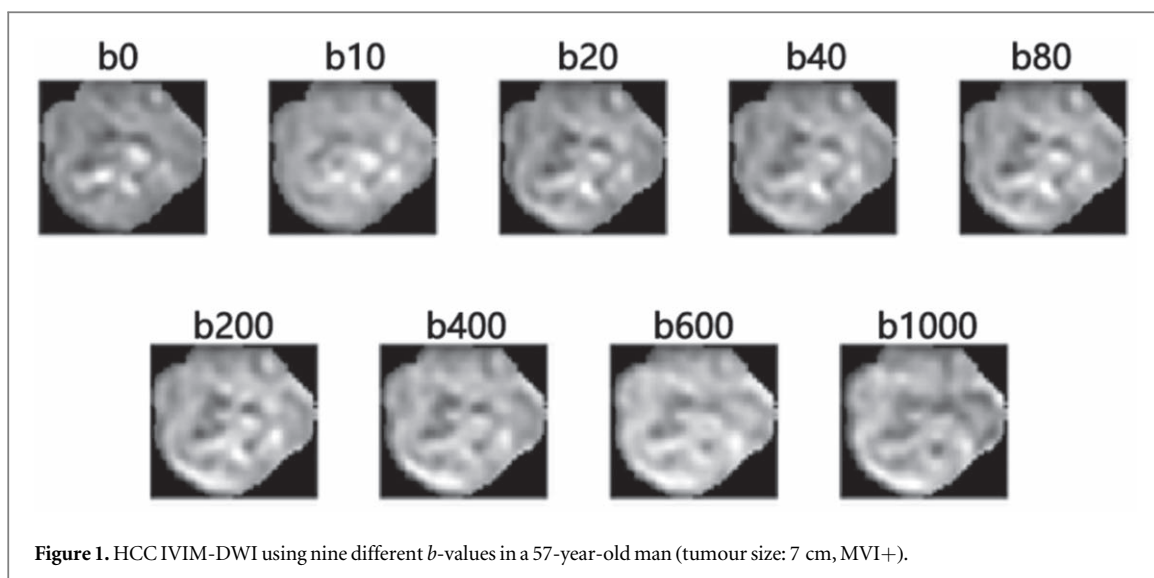


Figure 1. HCC IVIM-DWI using nine different b -values in a 57-year-old man (tumour size: 7 cm, MVI+).

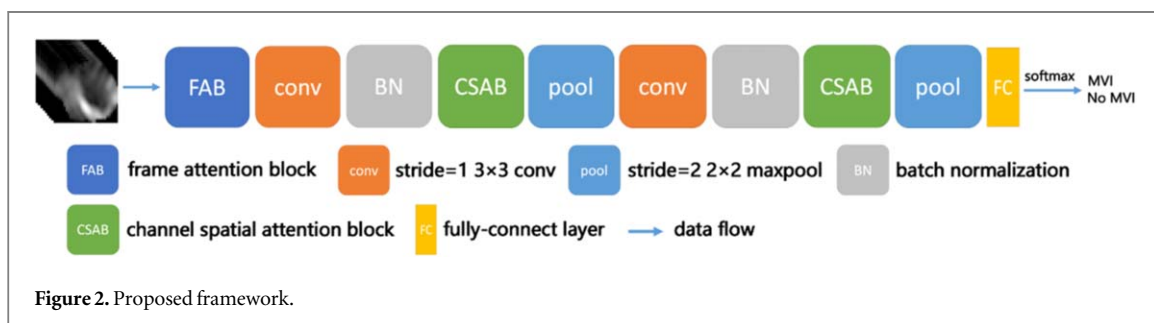


Figure 2. Proposed framework.

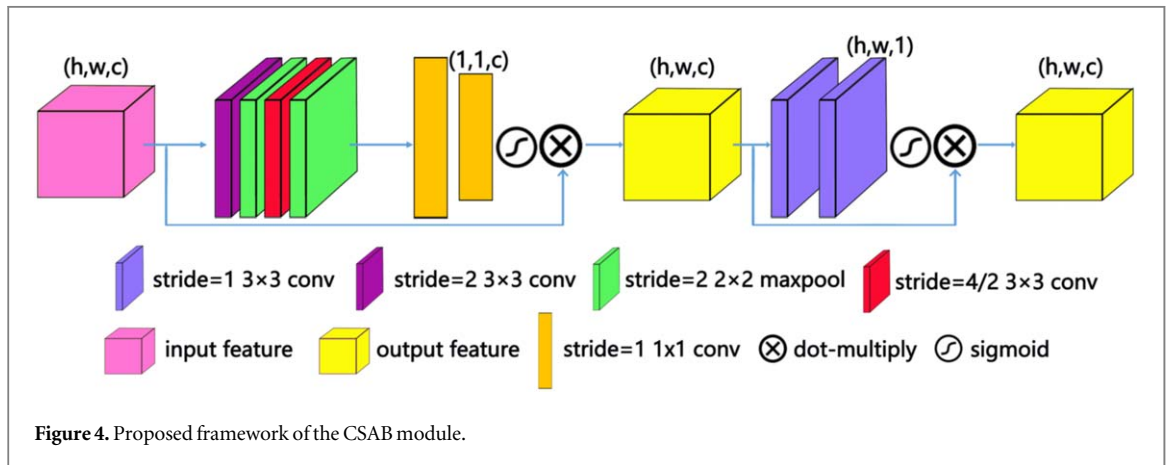
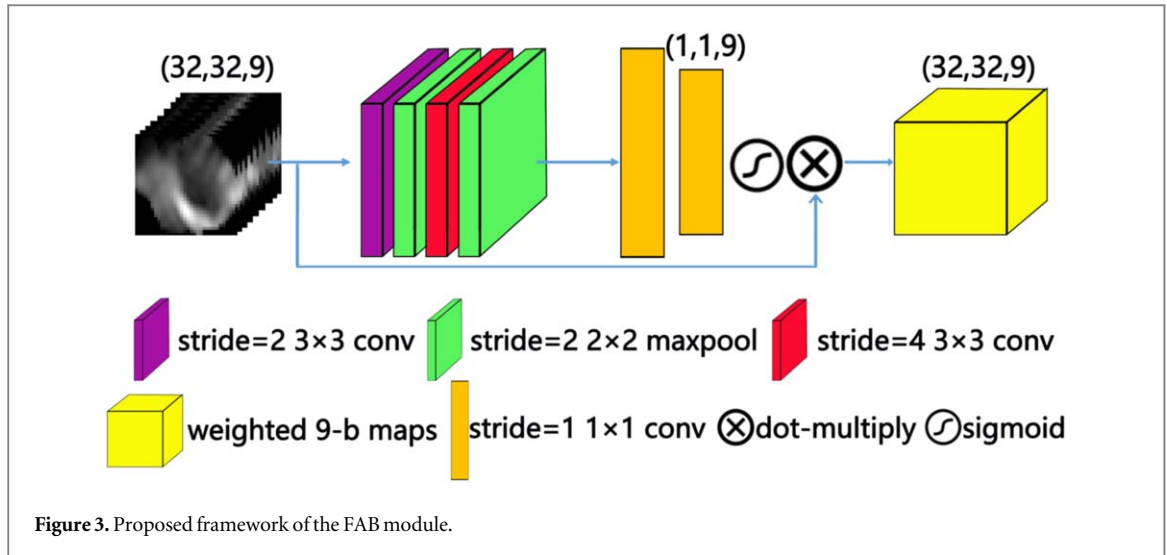
matrix, 5.5 mm slice thickness, field-of-view = 400 mm $\times$ 400 mm $\times$ 198 mm, slice gap = 0.5 mm, slices = 36, NSA = 2). IVIM-DWI employed respiratory-triggered single-shot echo planar imaging (EPI, EPI factor=53). Axial DWI was performed with the following nine b -values ($b = 0, 10, 20, 40, 80, 200, 400, 600$, and 1000 s mm^{-2}). Our 3.0-T MRI was equipped with a radiofrequency dual-transmission source system to guarantee the reliability of IVIM. Of the 98 lesions, 60 were pathologically confirmed to have an absence of MVI (MVI−), while 38 were confirmed as having MVI (MVI+).

2.2. Data preprocessing

A total of 98 HCC lesions were included in this study, of which 64 lesions were selected as the training set and 34 lesions were used as the test set. The 3D area of each lesion was manually segmented by an experienced radiologist using ITK-SNAP software. Figure 1 shows a representative IVIM-DWI of HCC obtained using nine different b -values in the axial view. The training set consisted of the slice with the largest cross-sectional area along the axial plane and the upper and lower slices of the slice with the largest area in the 3D region of the 64 lesions, while the test set consisted of one 2D slice with the largest cross-sectional area in the 3D region of the 34 lesions. Therefore, the training and test sets did not overlap. Subsequently, due to the different sizes of these tumours, we normalised the size of the axial slice of each tumour volume to 32×32 . In addition, because in each tumour only the slices with a large area are selected, and each tumour with IVIM-DWI have images with nine different b -values, the images with nine different b -values corresponding to each slice could be superimposed to form $32 \times 32 \times 9$ volumes. In this way, three $32 \times 32 \times 9$ volumes in the training set and one $32 \times 32 \times 9$ volume in the test set corresponding to the axial slices could be obtained for each tumour from the original MR slices. To increase the amount of data available to train the network, random rotation around the centre was adopted for data augmentation. Data augmentation was only conducted on the training dataset and not on the test dataset.

2.3. Proposed framework

As shown in figure 2, we proposed an end-to-end frame-channel-spatial attention network (FCSA-net) to directly extract spatiotemporal features from a series of IVIM-DWI with different values to predict MVI. The input of the network was a nine- b -value volume with a size of $32 \times 32 \times 9$, and the output of the network was the MVI prediction. Specifically, we introduced the attention mechanism to select dominant features from specific



b -value maps, channels and spatial areas of multiple b -value images to better characterise lesions. Two attention modules are proposed in the network: frame-attention-block (FAB) and channel/spatial attention block (CSAB). The FAB module was proposed to assign weights to different b -value maps, while the CSAB module was proposed to assign weights of deep features from different channels and spatial areas. We introduce the two proposed modules in detail in the following sections.

2.4. FAB module

The core of the FAB module extracts the weights of the different frames of the nine- b -value images. The structure of the FAB module is illustrated in figure 3. The input of the FAB is a 9- b image with shape (32, 32, 9), and the output was a weighted 9- b image with shape (32, 32, 9). We used the calculation method in equation (1) to obtain the b -value weight vector (BWV) for shape (1, 1, 9), and the BWV and 9- b image were multiplied as shown in equation (2) to obtain the assignment $output_1$. The weighted 9- b image was used as the output of the FAB

$$BWV = \sigma(FC_1(f_2(f_1(x)))), \quad (1)$$

$$output_1 = BWV \cdot x, \quad (2)$$

where σ represents the sigmoid activation function, x represents the input 9- b image, f_1 represents a 3×3 convolution with stride=2/filter=9/activation=relu and 2×2 max pooling layer with stride=2; f_2 represents 3×3 convolution with stride=4/filter=9/activation=relu and max pooling layer with stride=2; and FC_1 represents a 1×1 convolutional layer with stride=1/filter=18/activation=relu and 1×1 convolutional layer with stride=1/filter=9.

2.5. CSAB module

The core of the CSAB module extracts the channel/spatial weights of the deep features. The structure of the CSAB module is illustrated in figure 4. The CSAB module contains two components: a feature channel attention

module and feature spatial attention module. Assuming that the shape of the current input feature X is (h, w, c) , the former feature channel attention module is responsible for extracting the feature channel weight vector (FCWV) of shape $(1, 1, c)$, where c is the number of channels of the current feature, and the computation of FCWV is defined as equation (3). The dot multiplication operation is then used to perform the assignment to obtain $output_2$ as in equation (4); the latter feature spatial attention module is responsible for extracting the feature spatial weight matrix (FSWM) of shape $(h, w, 1)$, where h and w represent the length and width of the current feature, respectively. The FSWM was computed using equation (5), and the point multiplication operation was used to realise the weight assignment operation, as shown in equation (6), so that the weighted feature is obtained as $output_3$ of CSAB

$$FCWV = \sigma(FC_2(f_4(f_3(X)))), \quad (3)$$

$$output_2 = FCWV \cdot X, \quad (4)$$

$$FSWM = \sigma(f_5(output_2)), \quad (5)$$

$$output_3 = FSWM \cdot output_2, \quad (6)$$

where σ represents the sigmoid activation function, X represents the deep features of the current input of the CSAB; f_3 represents a 3×3 convolution with stride=2/filter= c /activation=relu and 2×2 maximum pooling layer with stride=2; f_4 represents 3×3 convolution with stride=4 or 2 (the first CSAB takes 4, the second CSAB takes 2)/filter= c /activation=relu and 2×2 maximum pooling layer with stride=2; f_5 represents 3×3 convolution with stride=1/filter= $c/2$ /activation=relu and 3×3 convolution with stride=1/filter=1; and FC_2 is a 1×1 convolution with stride=1/filter= 2^*c /activation=relu and 1×1 convolution with stride=1/filter= c .

2.6. Implementation

We used the open-source deep learning framework, Tensorflow, to implement the proposed framework and an RTX2080ti server graphics card. The loss function uses cross entropy, and we used the gradient descent method to minimise the loss function. The initial learning rate of the gradient descent method was $1e-3$, and the decay rate per 100 steps was 0.98. The batch size of each iteration was 64, and the number of training sessions was 40 000. To avoid the performance degradation caused by overfitting, we adopted an early stop strategy. The source code for the basic implementation is available at (<https://github.com/1057939502/attention-model-for-MVI-prediction-of-HCC>).

3. Experimental results

3.1. Training and test dataset

The dataset consisted of 98 lesions, of which 39 lesions were MVI− and 25 lesions were MVI+ were randomly selected to form the training set, while 21 MVI− lesions and 13 MVI+ lesions were included in the test set. The typical metrics of accuracy, sensitivity, specificity, receiver operating characteristic curve (ROC), and area under the curve (AUC) values were adopted to assess the performance of the proposed method for predicting MVI. To reduce the measurement error, training and testing was repeated 10 times to obtain mean±standard deviation to evaluate the performance of the model. To demonstrate the superiority of the proposed framework, we also implemented a 3D CNN that has no attention mechanism as the backbone for direct comparison. Furthermore, we also implemented a previously proposed deep feature fusion method with DWI for performance comparison, including concatenation (Dou et al 2018) and deep supervised net (DSN) (Lee et al 2014, Lessmann et al 2018, Zhou et al 2019, Wang et al 2020). Typical DWI contains three b -value images, and three CNNs were used separately to extract features from each of the three b -value images, followed by feature concatenation or DSN for lesion characterisation. Alternatively, four-fold cross validation with 10 repetitions was also conducted to assess the performance of different models for MVI prediction. Finally, visualisation of the weight vector and matrix of the proposed attention modules (BWV, FCWV, and FSWM) was conducted to demonstrate the effectiveness of the proposed method.

3.2. Performance comparison of different models for MVI prediction

We used the 3D CNN as the backbone to conduct the ablation experiments. As shown in table 1, using only the FAB module to assign weights to different b -values can bring performance improvements (AUC: 0.805 versus 0.8), and only using the CSAB module to assign weights to the channel/spatial deep features can also bring performance improvements (0.838 versus 0.8), while the performance is best (AUC: 0.85) when using the FAB and CSAB modules simultaneously, indicating the effectiveness of the proposed attention modules for improving MVI prediction. The loss curves, accuracy curves, and ROC curves of different models for MVI prediction are shown in figure 5.

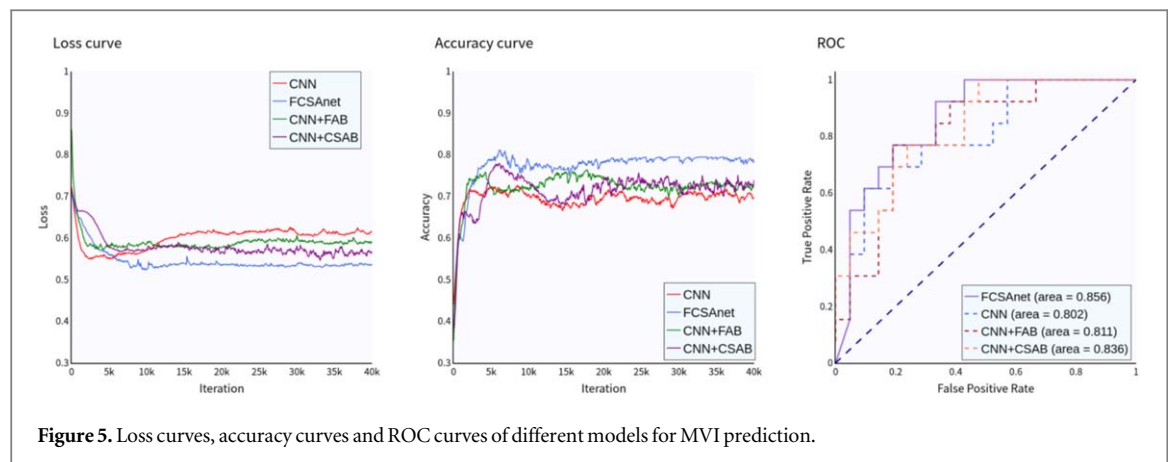


Figure 5. Loss curves, accuracy curves and ROC curves of different models for MVI prediction.

Table 1. Comparison of performance of different models for predicting MVI assessed using fixed training and test sets.

Method	Accuracy	Sensitivity	Specificity	AUC
CNN	0.717 ± 0.025	0.721 ± 0.097	0.716 ± 0.055	0.8 ± 0.022
CNN+FAB	0.731 ± 0.023	0.669 ± 0.037	0.781 ± 0.04	0.805 ± 0.047
CNN+CSAB	0.766 ± 0.022	0.761 ± 0.067	0.756 ± 0.033	0.838 ± 0.042
FCSA-net	0.78 ± 0.017	0.776 ± 0.044	0.801 ± 0.03	0.85 ± 0.02

Table 2. Performance comparison of different models for predicting MVI assessed using fixed training and test sets.

Method	Accuracy	Sensitivity	Specificity	AUC	Time
3-b concat	0.702 ± 0.022	0.585 ± 0.074	0.786 ± 0.064	0.667 ± 0.028	10 min
9-b concat	0.748 ± 0.027	0.646 ± 0.053	0.819 ± 0.043	0.781 ± 0.024	26 min
3-b concat (DSN)	0.711 ± 0.025	0.607 ± 0.081	0.751 ± 0.066	0.674 ± 0.03	11 min
9-b concat (DSN)	0.757 ± 0.016	0.684 ± 0.026	0.815 ± 0.044	0.79 ± 0.019	36 min
CNN	0.717 ± 0.025	0.721 ± 0.097	0.716 ± 0.055	0.8 ± 0.022	8 min
FCSA-net	0.78 ± 0.017	0.776 ± 0.044	0.801 ± 0.03	0.85 ± 0.02	14 min

3.3. IVIM-DWI versus DWI for MVI prediction

Table 2 shows that using nine CNN channels to extract features from nine b -value maps has better overall performance (AUC: 0.781) better than only using three b -values (AUC: 0.667), indicating that including more b -value images in the MVI prediction leads to performance improvement; however, the training time increases due to the large number of parameters of the nine CNNs. Furthermore, the DSN fusion method can slightly improve the characterisation performance compared with the feature concatenation method. By comparison, the CNN model extracts spatiotemporal features from the nine b -value images, although the accuracy (accuracy: 0.717) is slightly lower than that with 9- b concatenation (accuracy: 0.748), while the AUC (0.8) is higher than that of the 9- b concatenation (0.781) and the training time is shorter (8 min), indicating that the CNN model used in this study is more efficient. Finally, compared to the 9- b feature concatenation and DSN fusion methods, the proposed FCSA-net proposed in this study achieves the best overall performance with a relatively short training time. Figure 6 shows the detailed accuracy curves, loss curves, and ROC curves for these methods of MVI prediction. As shown in table 3, the performance of the different models was assessed using four-fold cross validation with 10 repetitions. The performance comparison also showed the advantages of the proposed method compared with that of existing methods.

3.4. Visualisation of the proposed model for MVI prediction

As shown in figure 7(a), the axial view of the tumour region is displayed for further MVI prediction. Visualisation using Grad-CAM (Selvaraju *et al* 2017), which is shown in figures 7(b) and (c), reflected the dominant features in specific spatial areas for MVI prediction using the CNN model and proposed FCSA-net. The CNN models focus on the tumour is relatively scattered, while the proposed FCSA-net using an attention mechanism focuses more on the main parts of the tumour, ignoring some weak borders and noisy backgrounds, which might be related to MVI information. Figure 7(d) shows the frame weight of the FAB module, which

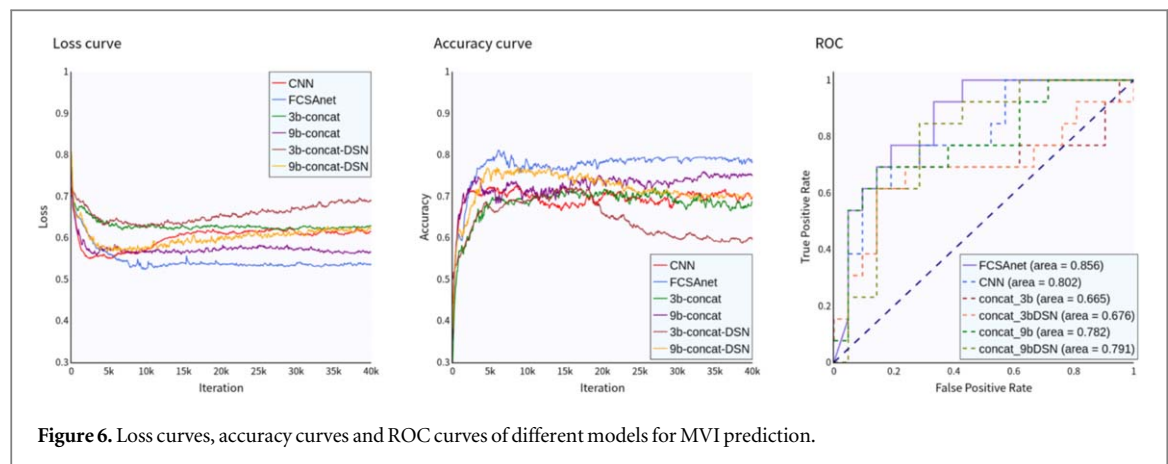


Table 3. Performance comparison of different models for MVI prediction in the training and test set using the four-fold cross validation with 10 repetitions.

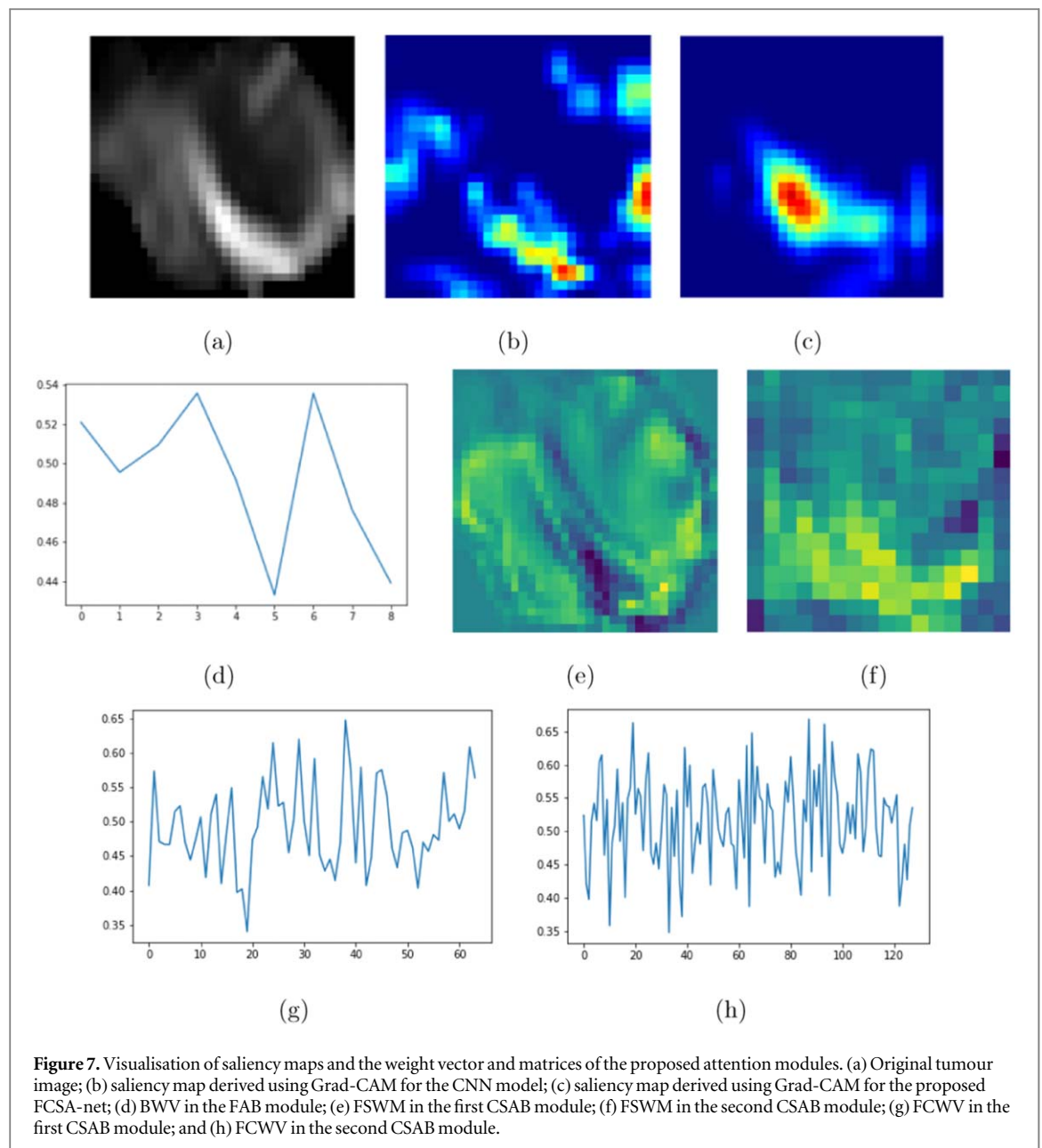
Method	Accuracy	Sensitivity	Specificity	AUC
3-b concat	0.679 ± 0.031	0.657 ± 0.069	0.702 ± 0.071	0.688 ± 0.052
9-b concat	0.742 ± 0.029	0.741 ± 0.061	0.712 ± 0.068	0.789 ± 0.048
3-b concat(DSN)	0.685 ± 0.021	0.671 ± 0.048	0.721 ± 0.048	0.711 ± 0.029
9-b concat(DSN)	0.751 ± 0.037	0.753 ± 0.077	0.728 ± 0.078	0.795 ± 0.059
CNN	0.696 ± 0.025	0.776 ± 0.057	0.632 ± 0.055	0.778 ± 0.034
CNN+FAB	0.711 ± 0.019	0.719 ± 0.039	0.689 ± 0.033	0.783 ± 0.021
CNN+CSAB	0.736 ± 0.035	0.781 ± 0.071	0.679 ± 0.072	0.818 ± 0.054
FCSA-net	0.766 ± 0.022	0.785 ± 0.054	0.738 ± 0.049	0.838 ± 0.035

reflects which of the nine b -values requires more attention. Figures 7(e) and (f) shows the corresponding spatial weights of the two CSAB modules, reflecting the parts that require more attention with respect to the deep features. Figures 7(g) and (h) reflects the channel weight of the two CSAB modules and indicates the channel that needs more attention with respect to the deep features.

4. Discussion

In this study, we proposed a deep learning network to extract spatiotemporal features directly from a series of b -value images of IVIM-DWI to predict MVI of HCC. Specifically, we introduced the attention mechanism to select dominant features from specific b -values, network channels, and spatial areas for MVI prediction. It should be noted that this study is the first to use deep learning with IVIM-DWI for lesion characterisation. Typical lesion characterisation using IVIM-DWI is based on the parameter values derived from the IVIM model (Li et al 2018, Zhao et al 2018, Wei et al 2019a, 2019b), in which the characterisation performance is considerably limited by noise and motion (Guyader et al 2015). Moreover, the reproducibility and reliability of IVIM parameters are also questionable in clinical practice (Iima and Le Bihan 2016). In contrast, the proposed deep learning method directly extracts the associated features that characterise MVI from a series of b -value images of IVIM-DWI in a data-driven manner, without generating IVIM parameter maps, and the representative performance of deep features has been widely demonstrated to be better than that of hand-crafted features (Litjens et al 2017, Zhou et al 2019). Therefore, the shortcomings of IVIM parameter estimation for lesion characterisation in clinical practice no longer exist in the proposed deep learning model.

A previous study attempted to use DWI with three b -value images (0, 100, 600) and deep learning to predict MVI of HCC, and the obtained characterisation performance had an AUC of 0.79 (Wang et al 2020). By comparison, the proposed deep learning model uses IVIM-DWI with nine b -values for predicting MVI of HCC and had much higher performance with an AUC of 0.85. We also assessed the performance of the reported deep learning model with fusion methods of feature concatenation (Dou et al 2018) and DSN (Lee et al 2014, Lessmann et al 2018, Zhou et al 2019) with nine b -value images, and the obtained AUC values were 0.781 and 0.79, respectively (table 2), indicating that the deep feature fusion method with concatenation or DSN is inferior to the proposed deep learning model with an attention mechanism. It should be noted that the major difference between the proposed deep learning model and the previous deep feature fusion method is that we extracted salient spatiotemporal features from nine b -value images using a 3D CNN with an attention mechanism,



whereas the deep feature fusion method uses CNN to extract deep features from each b -value image and then fuses the deep features from all b -value images for lesion characterisation. The superiority of the proposed model may be attributed to the extraction of salient spatiotemporal features between adjacent b -value images of the lesion, while the previous study did not consider this important information for lesion characterisation.

Attention mechanisms have been widely used in the field of deep learning (Vaswani *et al* 2017). Recently, Park *et al* studied the influence of an attention mechanism in a CNN (Park *et al* 2020). By designing a module called bottleneck attention (BAM) that is given an intermediate feature map, BAM can effectively follow two decomposition axes (channel and space) to generate attention maps, but the overhead is negligible. Research shows that BAM can significantly improve the performance of CNNs. In fact, the proposed FAB/CSAB module is quite different from CBAM in terms of extracting dimensionality reduction features. With CBAM it is very simple to extract the maximum/average value of the feature for dimensionality reduction (for example, a 32×32 feature map only retains its pixel with the highest grey value; a channel vector of 128 retains only the channel with the highest value); this simple and crude dimensionality reduction method leads to the loss of a lot of effective contextual information regarding deep features, and the parameter of CBAMs attention pathway is very small, so that the acquired weight may not be the most effective or stable. The proposed FAB/CSAB module takes this into consideration and expands the receptive field to retain the effective information from deep features as much as possible by connecting multiple convolution layers and downsampling layer by layer, which has been demonstrated to be effective in encoding contextual information in Gather-Excite network for visual tasks (Hu *et al* 2018). The experimental results of the ablation study (table 1) verified the effectiveness of the

proposed attention modules for improving characterisation performance. Meanwhile, removing features that are not closely related to MVI through the attention mechanism (figures 7(b) and (c)) remarkably improves the characterisation performance. We believe that a more advanced attention mechanism may further improve the performance of lesion characterisation based on IVIM-DWI.

At present, previous studies have not yet performed clinical interpretation of IVIM for lesion characterisation using deep learning networks. A clear understanding and interpretation of the relationship between IVIM b -value images and pathological information is essential for the clinical application of deep networks. In the present study, by visualising the weight vector and matrix of the attention mechanism (figure 7(d) to (h)), we can observe which information in IVIM-DWI determines the characterisation performance. If such salient information can be used to identify the key b -value regions that determine the characteristics of the lesion, it will help to clinically determine the optimal b -value sequence for imaging and diagnosis of specific lesions. In addition, the visualisation results (figures 7(d) to (h)) also verified our hypothesis, in which not all b -value images and all the extracted deep features were helpful for lesion characterisation. We believe that clinical interpretability by visualising the weight matrix should be of great significance for the diagnosis of clinical diseases in the future.

In addition, the current clinical routine DWI scans in hospitals are based on three b -value images, which are mainly used for the detection of lesions and the fitting generation of ADC parameter maps for lesion characterisation (Koh and Collins 2007, Taouli and Koh 2010, Shenoy-Bhangle *et al* 2017). As a new trend in DWI, current IVIM-DWI approaches need to use more b -values to accurately fit the IVIM model to obtain D , D^* , and f parameter maps for lesion characterisation. The proposed method shows that based on the current multiple b -value images of IVIM-DWI, mining potential multiple b -value image sequence information using deep networks can obtain better MVI prediction performance (tables 2 and 3) than using the three- b -value image method proposed in previous studies (Zhou *et al* 2019, Wang *et al* 2020). Moreover, this process directly makes predictions based on IVIM-DWI images without relying on the parameter fitting of traditional IVIM-DWI models. Nonetheless, the acquisition of more b -value images requires more scanning time. If the purpose of the clinical routine is to detect lesions, the three b -value images of routine DWI are sufficient. However, from the perspective of lesion characterisation, it is worthwhile to collect more b -value images to improve tumour characterisation performance.

The present study has several limitations. First, the method we proposed was only verified on data from a single clinical centre, the amount of data is relatively small, and the scanning parameters were relatively fixed. The performance measurement of this model in multicentre, large sample, and different model scanning conditions will be important to examine in the future. Furthermore, the visualised weight matrix can reveal the imaging modality and regional information associated with pathological tumour information, but whether the regional information determined using a computer in a data-driven manner can help clinical diagnosis is debatable. In the future, through close cooperation with clinicians, the combination of weighted visualisation information and clinical experience may further improve tumour characterisation performance.

5. Conclusion

In this study, we proposed an attention-based deep learning model to directly extract spatiotemporal features from IVIM-DWI b -value image series for MVI prediction of HCC. The qualitative and quantitative experimental results verified the excellent performance of this method compared to that of previous methods. In addition, the attention mechanism selects dominant features from b -values, channels, and spatial regions, which can achieve better characterisation of lesions. We hope that this model can be widely used for clinical tumour characterisation based on IVIM-DWI.

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